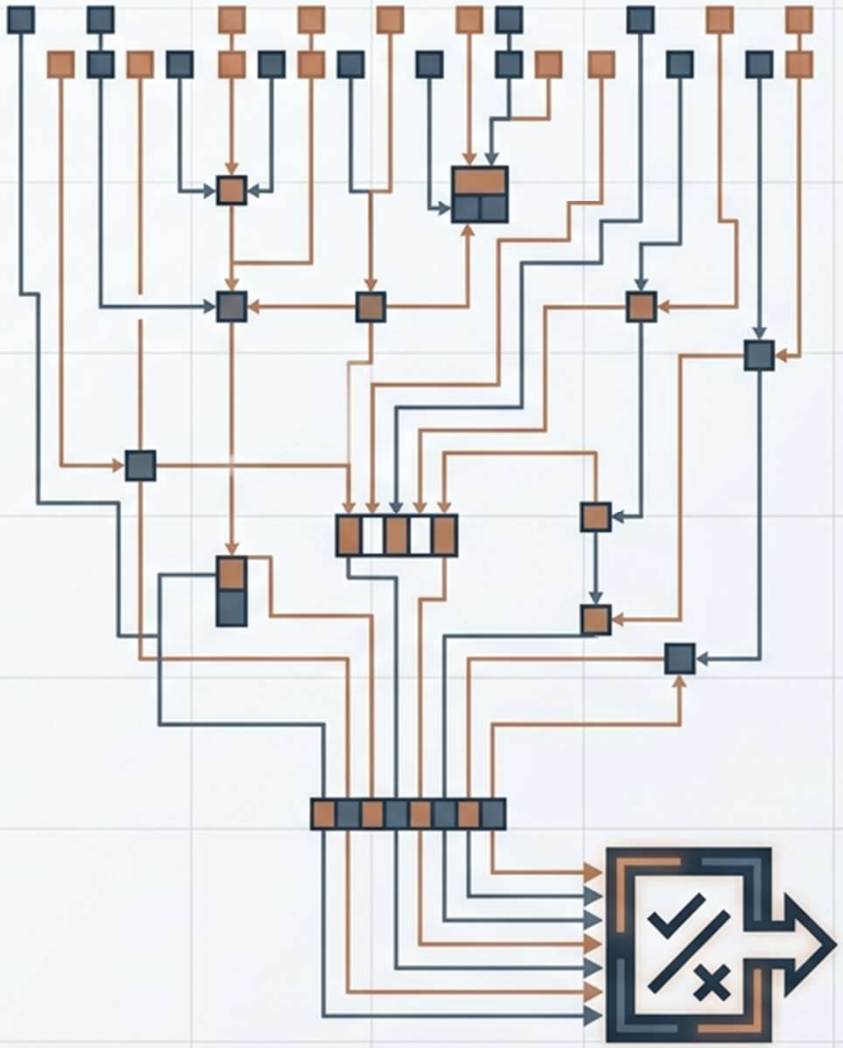
Automating the York City Bank Decision Engine

An end-to-end machine learning pipeline to instantly assess home loan eligibility across urban, semi-urban, and rural markets.



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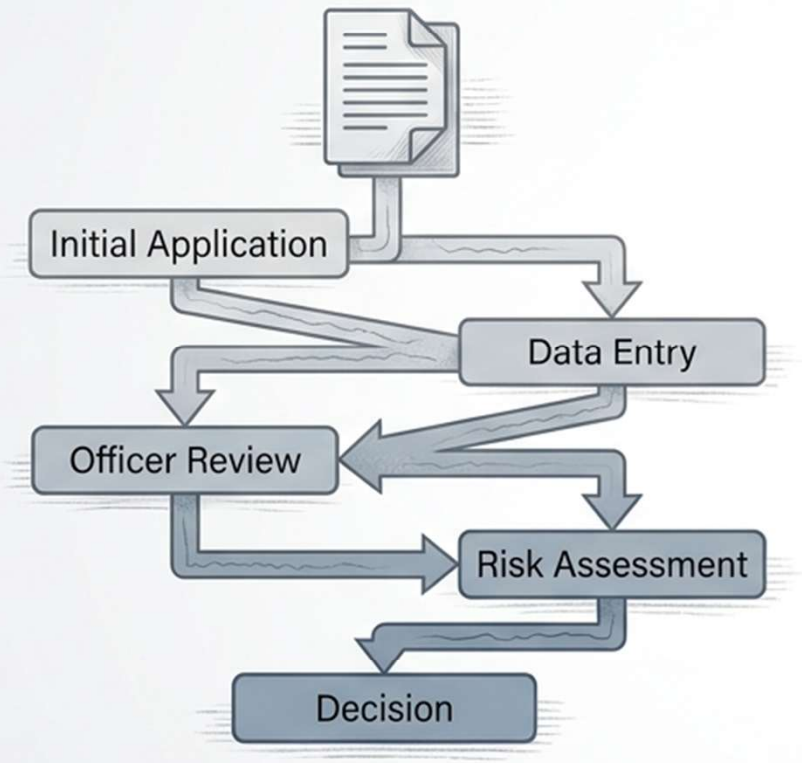
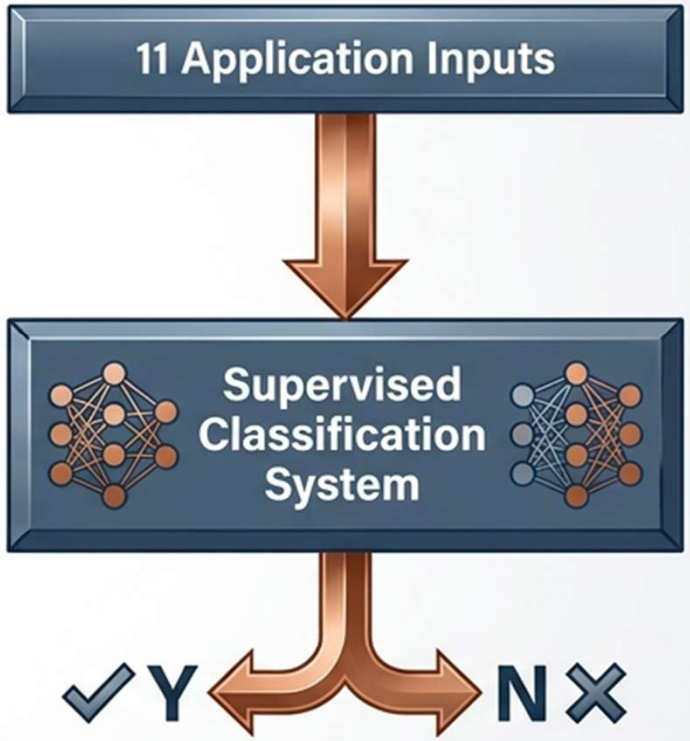
Transitioning from manual reviews to instant supervised classification





YCB's objective is to accurately identify eligible customer segments at the point of application, allowing targeted home loan originations without human review latency.

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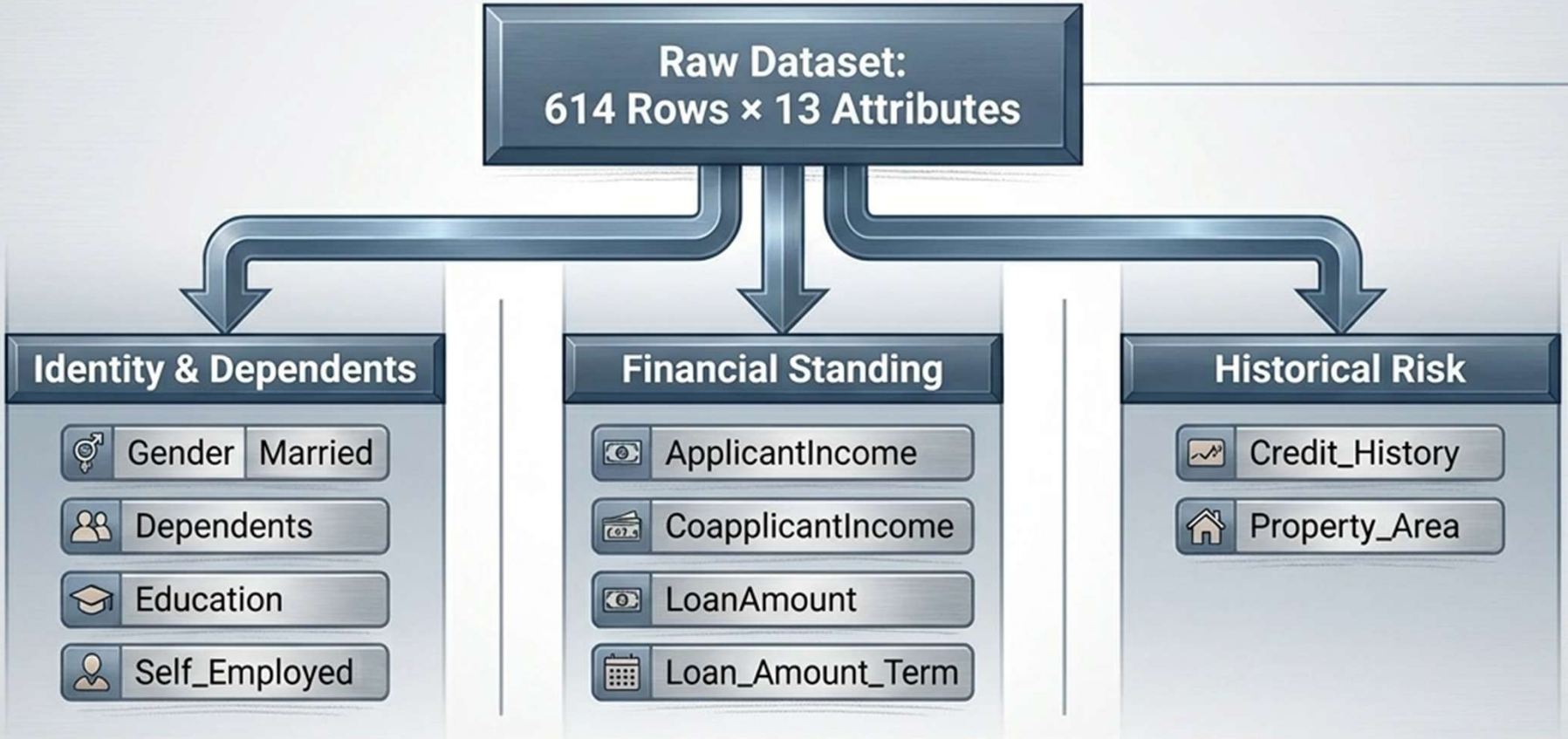
Deconstructing the 614 applicant profiles into three core dimensions



The pipeline drops the unique **Loan_ID** as non-predictive, leaving a concise, 11-feature input framework for the prediction target: **Loan_Status**.

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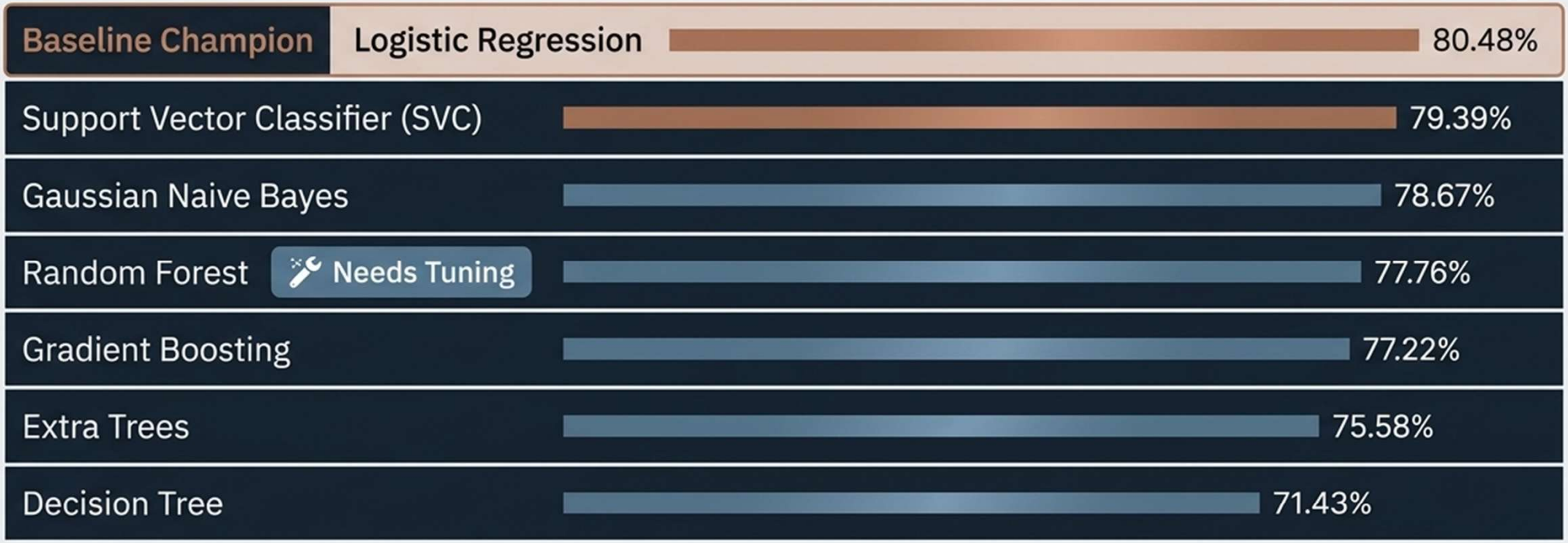
Baseline shootout identifies Logistic Regression and SVC as early frontrunners



Complex ensemble methods like Random Forest initially lag behind simpler linear models due to an unconstrained hyperparameter architecture.

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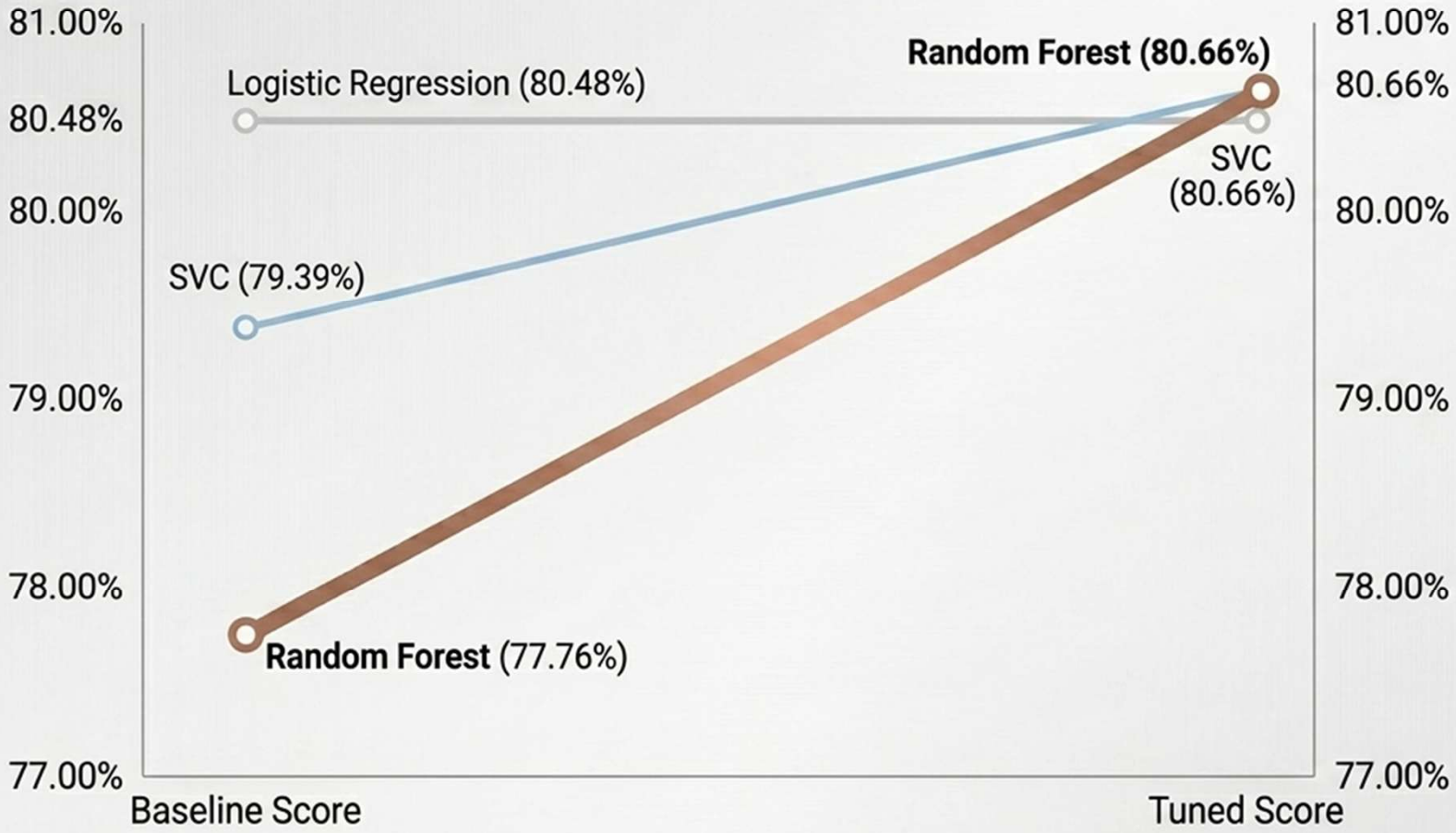
Randomized parameter tuning unlocks the Random Forest architecture



By systematically testing hundreds of parameter combinations using **RandomizedSearchCV**, Random Forest achieved a massive leap in accuracy, matching the absolute top score while retaining the structural robustness of a decision-tree ensemble.

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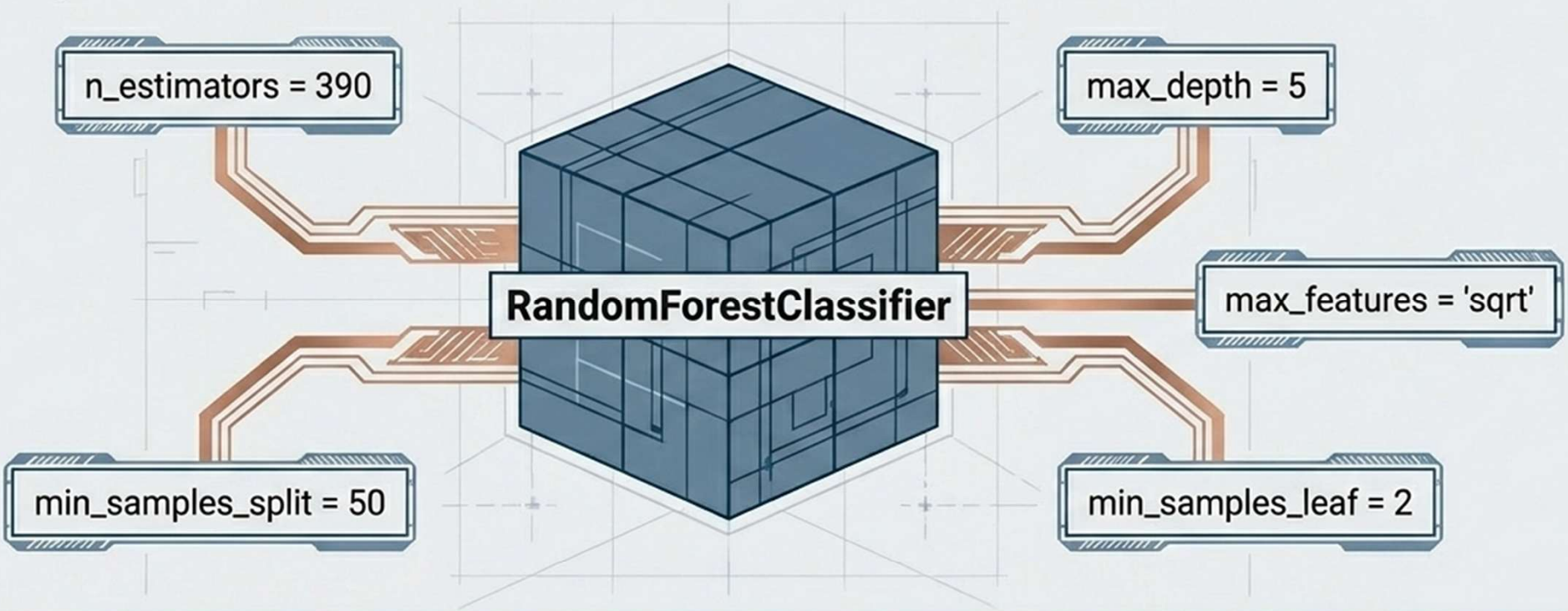
The finalized champion model is serialized and ready for deployment



The model matrix is fully trained on the unified dataset and serialized to disk via joblib as `loan_status_prediction`, freezing the intelligence for external integration.

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A simple interface connecting loan officers to the decision engine

Abstract algorithmic complexity is completely hidden from the user. YCB officers input **11 variables** and receive **millisecond eligibility routing,** eliminating the manual bottleneck entirely.

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